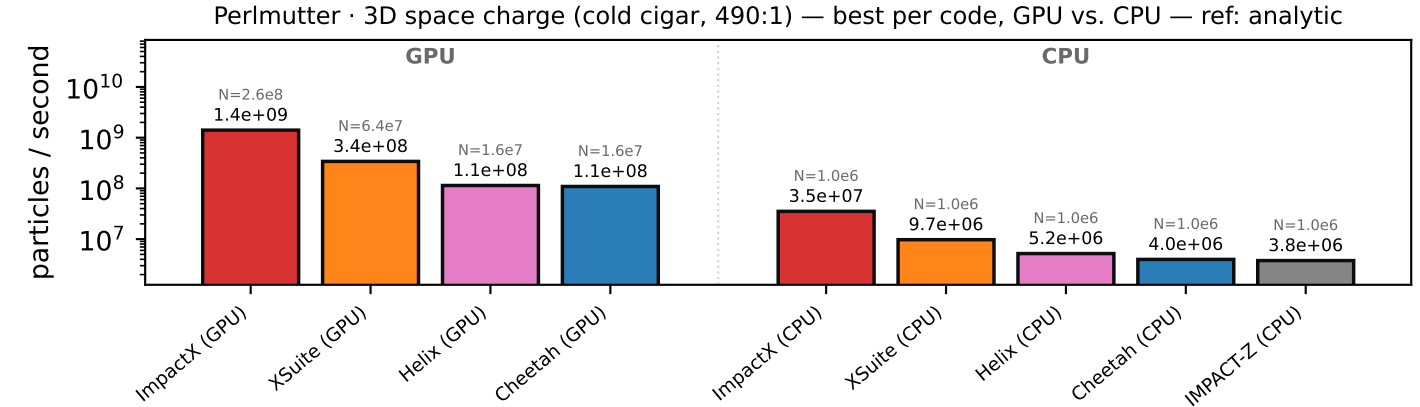

not shown, physics out of tolerance: PyOrbit (CPU), BMad (CPU), Synergia (GPU), Synergia (CPU)

kernel over a mesh cell: IGF = integrated in 3D; z-integrated = along z only, point-sampled in x,y; point 1/r = sampled at cell centres

each bar = the fastest config for that code on that device (any precision / fast-math); N = the particle count that bar was measured at (top of each device's sweep)

hardware: CPU: AMD EPYC 7763 (64 cores) GPU: NVIDIA A100-SXM4-40GB

versions: impactx 26.08 · xsuite 0.53.4 · helix v1.10.1 · cheetah 0.8.2 · impactz V2.7.7